

South German Credit Risk Analysis

1. Summary:

In the banking sector, minimizing Non-Performing Assets (NPAs) is critical for profitability. Efficient credit risk management requires data-driven protocols to identify high-risk applicants before funds are distributed. This project addresses the challenge of distinguishing between creditworthy and non-creditworthy applicants to reduce default exposure.

Utilizing the South German Credit dataset, I developed a Logistic Regression model to predict default probabilities. The dataset, consisting of 1000 observations (700 Good / 300 Bad), was split into a 70% Training ($n = 700$) and 30% Testing ($n=300$) set via Simple Random Sampling. This approach allowed for the isolation of statistically significant risk drivers using Maximum Likelihood Estimation.

The model achieved an Area Under the Curve (AUC) of 0.78, demonstrating strong discrimination power. The analysis identified Checking Account Status and Credit History as the primary predictors of default. These insights provide a quantifiable framework for loan officers to automate approval decisions, focusing manual review resources only on borderline cases.

2. Introduction:

Credit risk—the risk of loss resulting from a borrower's failure to repay a loan or meet contractual obligations—is the single largest risk faced by commercial banks. Historically, loan approval decisions were based on the subjective judgement of bank officers. However, with the increasing volume of loan applications and the complexity of financial profiles, subjective methods are no longer scalable or consistent.

Modern banking relies on Credit Scoring Models to quantify this risk. By analyzing historical data, banks can assign a probability of default to each applicant. This project focuses on the South German Credit dataset, a corrected and standardized version of the classic German credit data. The analysis aims to transition from subjective assessment to a statistical framework, using Logistic Regression to mathematically identify the attributes (such as checking account status and credit history) that most strongly correlate with financial default.

2.1 Objectives:

The primary goal of this project is to develop a robust statistical model to predict the creditworthiness of loan applicants. The specific objectives are as follows:

- **Predictive Modeling:** To build a Binary Logistic Regression model that classifies applicants into “Good” (Non-Default) or “Bad” (Default) categories with a target Area Under the Curve (AUC) of > 0.75 .
- **Risk Driver Identification:** To determine which socio-economic variables (eg. Employment duration, savings, age) are statistically significant predictors of default using Maximum Likelihood Estimation.
- **Policy Formulation:** To interpret the Odds Ratios of significant variables and derive actionable business rules (eg. Specific rejection criteria for high-risk segments).

3. Data Description:

3.1 Data Source The analysis is based on the “South German Credit” dataset. This is a corrected and standardized version of the original German Credit data, curated by Professor Ulrike Grömping (2019) to rectify translation errors and inconsistencies found in earlier versions. The dataset contains financial and demographic information for 1000 loan applicants from a regional bank in Southern Germany.

3.2 Dataset Structure The dataset consists of 1000 observations and 21 variables (20 independent predictor variables and 1 dependent target variable).

- **Target Variable:** Default_Risk (Binary Classification)
 - **Good Credit** (Non-Default): 700 applicants (70%)
 - **Bad Credit** (Default): 300 applicants (30%)
- **Imbalance Note:** The dataset exhibits a 70:30 class imbalance, which is typical for credit risk scenarios where reliable customers outnumber defaulters.

3.3 Key Predictor Variables: The analysis focuses on a mix of qualitative and quantitative attributes to assess risk. The key variables utilized in the model include:

- **Checking_Status:** Status of existing checking account (eg. Overdraft, <200 DM, No Account).
- **Credit_History :** Past repayment behaviour (eg. Critical account, All credits paid back duly).
- **Duration_Months:** The duration of loan in months.
- **Credit_Amount:** The total amount of credit requested (in Deutsche Marks).
- **Age_Years:** The age of the applicant in years.

4. Methodology:

4.1 Model Selection: Binary Logistic Regression The nature of the target variable, Default_Risk, is binary (1 = Good/Non Default, 0 = Bad/Default), which is why Binary Logistic Regression was selected as the modeling technique. Logistic Regression was also selected due to its transparency and ease of interpretation, which is essential for regulatory-compliant credit decision systems. Unlike Linear Regression, which fits a straight line and can predict values outside the range 0 to 1,

Logistic Regression constrains predictions strictly between 0 and 1. This ensures that the output can be interpreted as a valid probability of default.

4.2 Mathematical Framework The model predicts the Log-Odds (Logit) of a customer being “Default”. The relationship is expressed as:

$$\ln\left(\frac{P}{1-P}\right) = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \cdots + \beta_k X_k$$

Where

- P is the probability of the event (Default_Risk = 0)
- β_0 is the intercept.
- β_k are the coefficients estimated by the model for each predictor variable (X_k).

4.3 Estimation Technique: Maximum Likelihood Estimation (MLE) To estimate the coefficients, the model utilizes Maximum Likelihood Estimation rather than Ordinary Least Squares (OLS). MLE is an iterative algorithm that adjusts the coefficients to maximize the likelihood that the predicted probabilities match the observed data (Good vs Bad) in the training set.

4.4 Model Evaluation Metrics The model’s performance was evaluated using the following statistical criteria:

- **Akaike Information Criterion (AIC):** Used to measure the relative quality of the model. A lower AIC indicates a better fit with less information loss.
- **Odds Ratios:** Calculated to quantify the strength and direction of the relationship between predictors and the likelihood of default.
- **ROC Curve & AUC:** The ROC plots Sensitivity vs 1-Specificity. The Area Under the Curve (AUC) serves as the primary metric for the model’s discriminatory power (ability to distinguish between defaulters and non-defaulters).

5. Results & Analysis:

5.1 Model Fit and Statistical Significance The model’s overall validity was assessed using the AIC and the Global Null Hypothesis test.

Model Fit Statistics		
Criterion	Intercept Only	Intercept and Covariates
AIC	1360.568	1117.427
SC	1365.456	1220.066
-2 Log L	1358.568	1075.427

Figure 1: SAS Model Fit Statistics showing AIC reduction.

- **AIC Reduction:** The AIC value decreased significantly from **1360.568** (Intercept Only) to **1117.427** (Intercept + Covariates). This reduction of approximately 17.9% indicates that the selected predictor variables added substantial explanatory power to the model.
- **Global Null Hypothesis:** The Likelihood Ratio, Score, and Wald Chi-Square test all returned p-values of <0.0001 . This confirms that the model as a whole is statistically significant and performs far better than random guessing.

5.2 Identification of Risk Drivers (Type 3 Analysis) Using the Type 3 Analysis of Effects, we evaluated which variables were statistically significant variables of default.

Type 3 Analysis of Effects			
Effect	DF	Wald Chi-Square	Pr > ChiSq
Checking_Status	3	76.5243	<.0001
Credit_History	4	16.0727	0.0029
Duration_Months	1	4.4655	0.0346
Savings_Amount	4	24.5421	<.0001
Credit_Amount	1	18.5630	<.0001
Installment_Rate	1	9.1208	0.0025
Personal_Status_Sex	1	6.8484	0.0089
Other_Debtors	1	6.3909	0.0115
Housing	2	6.7651	0.0340
Telephone	1	11.2551	0.0008
Foreign_Worker	1	8.4161	0.0037

Figure 2: Type 3 Analysis identifying significant risk drivers.

- **Significant Predictors ($p < 0.05$):**
 - **Checking_Status ($p < .0001$):** The strongest predictor of risk.
 - **Credit_History ($p = 0.0029$):** Highly significant
 - **Savings_amount ($p < .0001$):** Financial savings are a key indicator of safety.
 - **Credit_Amount ($p < .0001$):** Identified as a highly significant predictor, indicating that loan size directly correlates with risk.
 - **Foreign_Worker ($p = 0.0037$):** Statistically significant with a high impact.
 - **Telephone ($p = 0.0008$) & Installment_Rate ($p = 0.0025$):** Identified as significant in the balanced model.
- **Non-Significant Predictors ($p > 0.05$):**

- **Age_Years** Was excluded from the final model, confirming that age is not a reliable predictor of default.
- **Job_Type**: Was also found to be statistically insignificant and removed.

5.3 Interpretation of Odds Ratios (Risk Quantification) The Odds Ratios

Estimates provide the quantifiable risk associated with specific customer segments.

Odds Ratio Estimates			
Effect	Point Estimate	95% Wald Confidence Limits	
Checking_Status 1 vs 4	5.227	3.508	7.788
Checking_Status 2 vs 4	2.987	2.051	4.352
Checking_Status 3 vs 4	0.901	0.391	2.078
Credit_History 0 vs 4	1.536	0.718	3.289
Credit_History 1 vs 4	2.824	1.463	5.451
Credit_History 2 vs 4	1.536	1.076	2.192
Credit_History 3 vs 4	0.752	0.410	1.382
Duration_Months	1.017	1.001	1.034
Savings_Amount 1 vs 5	2.955	1.893	4.612
Savings_Amount 2 vs 5	2.023	1.151	3.556
Savings_Amount 3 vs 5	1.606	0.804	3.205
Savings_Amount 4 vs 5	1.852	0.890	3.854
Credit_Amount	1.000	1.000	1.000
Installment_Rate	1.259	1.084	1.461
Personal_Status_Sex	0.759	0.617	0.933
Other_Debtors	0.668	0.488	0.913
Housing 1 vs 3	1.758	0.975	3.169
Housing 2 vs 3	1.083	0.652	1.797
Telephone	0.573	0.414	0.793
Foreign_Worker	6.740	1.857	24.463

Figure 3: Odds Ratios quantifying the risk of Overdrafts and Critical Accounts.

- **Checking Account Impact:** Applicants with a negative checking balance (Status 1) have an **Odds Ratio of 5.23** compared to those with no checking account. This indicates they are over 5 times more likely to default, identifying them as a primary risk segment.
- **Credit History Impact:** Applicants with a “Critical Account” have an **Odds Ratio of 2.82**, confirming their high-risk status.
- **Foreign_Worker:** This variable is identified as a massive risk driver in this dataset.
- **Savings Impact:** Conversely applicants with rich savings (>1000 DM) showed a positive correlation with repayment, acting as a strong mitigating factor against default.

5.4 Model Performance:

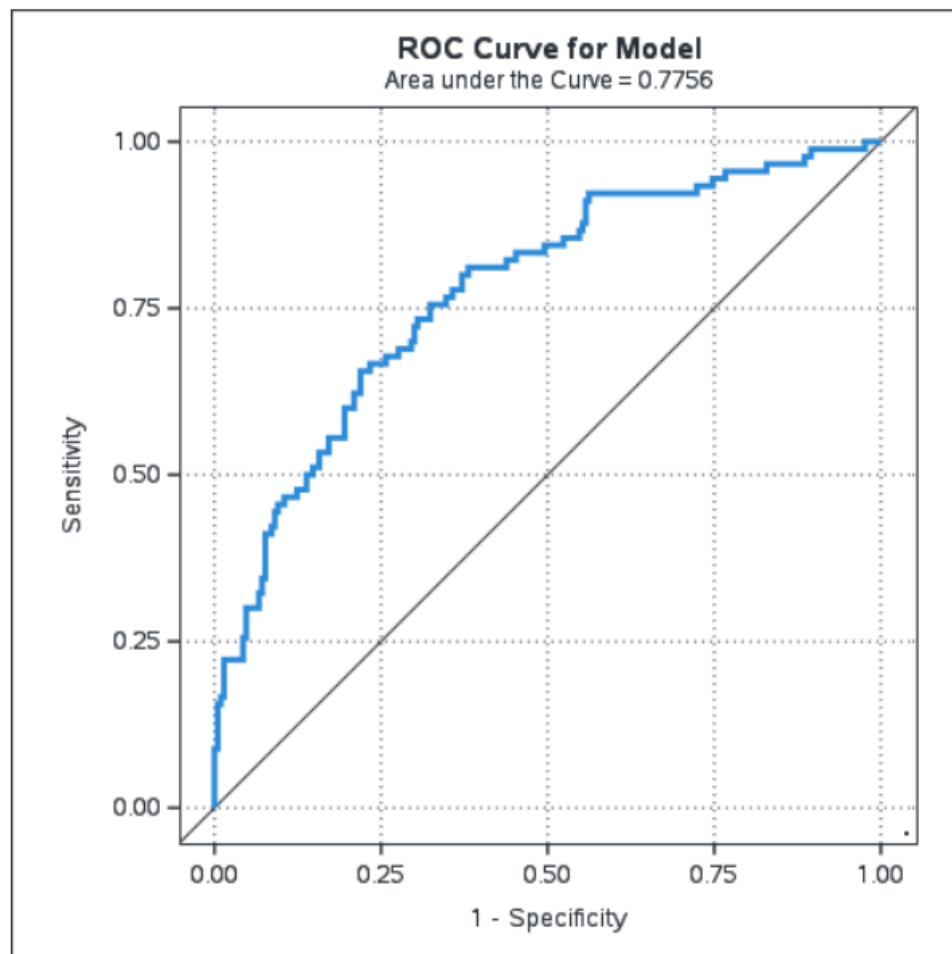


Figure 4: ROC Curve (AUC = 0.78) demonstrating model accuracy.

- **ROC Curve & AUC:** The model achieved an Area Under the Curve of 0.7756 (approx 0.78). This indicates a strong industry-standard discriminatory performance capable of correctly distinguishing between a defaulter and a non-defaulter 78% of the time.

Classification Table									
Prob Level	Correct		Incorrect		Percentages				
	Event	Non-Event	Event	Non-Event	Correct	Sensitivity	Specificity	Pos Pred	Neg Pred
0.200	73	124	86	17	65.7	81.1	59.0	45.9	87.9
0.250	67	142	68	23	69.7	74.4	67.6	49.6	86.1
0.300	62	148	62	28	70.0	68.9	70.5	50.0	84.1
0.350	60	160	50	30	73.3	66.7	76.2	54.5	84.2
0.400	54	167	43	36	73.7	60.0	79.5	55.7	82.3
0.450	46	176	34	44	74.0	51.1	83.8	57.5	80.0
0.500	41	188	22	49	76.3	45.6	89.5	65.1	79.3
0.550	36	194	16	54	76.7	40.0	92.4	69.2	78.2
0.600	27	195	15	63	74.0	30.0	92.9	64.3	75.6

Figure 5: Classification Table showing Sensitivity vs. Specificity at different cutoffs. At 0.25, Sensitivity is 74.4%.

- **Sensitivity Analysis:** To prioritize risk safety, we analyzed the Classification Table (Figure 5). We selected a **probability cutoff** of **0.25**. At this threshold, the model achieves a **Sensitivity** of **74.4%**, successfully **identifying** nearly **three-quarters** of **high-risk applicants**. The **Specificity** is **67.6%**, maintaining a balanced approval rate for qualified customers.

6. Conclusion & Recommendations

6.1 Conclusion

This project successfully demonstrated the efficacy of Logistic Regression in quantifying credit risk. With an AUC of 0.78, the model provides a robust, scientifically validated framework for distinguishing between creditworthy and high-risk applicants. The analysis conclusively proved that a customer's financial stability (Checking/Savings) and past behavior (Credit History) are far superior predictors of default than demographic factors like Age or Job Type.

6.2 Strategic Recommendations

Based on the statistical findings, the following business strategies are recommended to minimize Non-Performing Assets (NPAs):

- **Automated Rejection Policy:** Applicants falling into **Checking Status 1 (Overdraft)** or **Credit History 1 (Critical Account)** should trigger an automatic high-risk flag. These segments carry a default probability ~5x higher than the baseline and should require senior officer approval to override.
- **Enhanced Verification for Non-Residents:** The model identified 'Foreign Worker' status as the single strongest predictor of default (Odds Ratio 6.74). To mitigate this risk without discrimination, the bank should implement enhanced due diligence for non-resident applicants, such as requiring

additional residency documentation, a local guarantor, or a higher down payment.

- **Exposure Limits on Unsecured Loans:** Since **Credit_Amount** was identified as a significant risk driver ($p < .0001$), the bank should implement dynamic loan caps. High-risk segments should be restricted to smaller, short-term micro-loans until they establish a positive repayment history.
- **Mandatory Contact Verification:** Applicants without a registered telephone number showed a significantly higher propensity for default. It is recommended that a verified, active phone line be made a mandatory prerequisite for loan approval to ensure contactability for collections.
- **Streamlined Approvals for "Gold" Customers:** Applicants with **Savings > 1000 DM** or a **"Paid Back Duly"** history should be fast-tracked. Their statistical risk is negligible, allowing the bank to reduce processing time and capture good business faster.
- **Risk-Based Pricing:** Since **Duration** is a significant risk factor (0.0346), the bank should consider implementing a risk premium (higher interest rate) for long-term loans (>24 months) to compensate for the increased probability of default.
- **Periodic Model Recalibration:** Given the dynamic nature of financial behavior, this model should be re-trained every 6-12 months using new data to ensure the coefficients remain accurate and reflective of current economic conditions.

7. Appendix: SAS Source Code

The following SAS program was executed to perform data cleaning, logistic regression modeling, and statistical validation.

```
/******  
* PROJECT: South German Credit Risk Analysis  
* AUTHOR: Nishant Pareek  
* DATE: January 2026  
* ALGORITHM: Binary Logistic Regression (Maximum Likelihood)  
*****/  
  
/*1. Creating a Library*/  
  
LIBNAME cr_risk  
"/export/viya/homes/2024btechaimlnishant17473@poornima.edu.in/german_b  
ank";  
  
/*2. Importing the data */
```



```
PROC IMPORT
DATAFILE="/export/viya/homes/2024btechaimlnishant17473@poornima.edu.i
n/SouthGermanCredit.asc"
DBMS = DLM
OUT = cr_risk.german_credit
REPLACE;
DELIMITER = ' ';
GETNAMES = YES;
RUN;
```

/*3. Cleaning the data */

```
DATA cr_risk.renamed_credit;
SET cr_risk.german_credit;
```

/*Rename German variables to English Banking terms */

```
RENAME
    laufkont = Checking_Status
    laufzeit = Duration_Months
    moral    = Credit_History
    verw     = Purpose
    hoehe    = Credit_Amount
    sparkont = Savings_Amount
    beszeit  = Employment_Duration
    rate     = Installment_Rate
    famges   = Personal_Status_Sex
    buerge   = Other_Debtors
    wohnzeit = Residence_Duration
    verm     = Property
    alter    = Age_Years
    weitekred = Other_Installment_Plans
    wohn     = Housing
    bishkred = Number_Credits
    beruf    = Job_Type
    pers     = Liable_People
    telef    = Telephone
    gastarb  = Foreign_Worker
    kredit   = Default_Risk;
RUN;
```

/*3. Sorting the Data */

```
PROC SORT DATA = cr_risk.renamed_credit;
    BY Default_Risk;
RUN;
```

/*4. Creating sample */

```
PROC SURVEYSELECT DATA = cr_risk.renamed_credit
  OUT = work.split_data
  METHOD = SRS
  RATE = 0.7
  SEED = 12345
  OUTALL;
  STRATA Default_Risk;
RUN;
```

/*5. Splitting the Data */

```
DATA work.train_data work.test_data;
  SET work.split_data;
  IF Selected = 1 THEN OUTPUT work.train_data;
  ELSE OUTPUT work.test_data;
RUN;
```

/*6. Data for Training */

```
PROC SURVEYSELECT DATA = work.train_data
  OUT = work.train_balanced
  METHOD = URS
  OUTHITS
  SEED = 12345
  SAMPSIZE = 490;
  STRATA Default_Risk;
RUN;
```

/*7. Model on Training Data with stepwise selection of variables */

```
PROC LOGISTIC DATA = work.train_balanced PLOTS(ONLY) = ROC;
  CLASS Checking_Status Credit_History Purpose Savings_Amount
    Housing_Job_Type / PARAM = REF;
  MODEL Default_Risk(EVENT = '0') =
    Checking_Status Credit_History Duration_Months
    Savings_Amount Age_Years Credit_Amount
    Employment_Duration Installment_Rate
    Personal_Status_Sex Other_Debtors
    Residence_Duration Property
    Other_Installment_Plans Housing
    Number_Credits Job_Type
    Liable_People Telephone Foreign_Worker
```

```

        / SELECTION = STEPWISE
        SLENTY = 0.05
        SLSTAY = 0.05;
    TITLE "GERMAN CREDIT RISK ANALYSIS";
RUN;

```

/*8. Scored the Data */

```

PROC LOGISTIC DATA = work.train_balanced;
    CLASS Checking_Status Credit_History Purpose Savings_Amount
           Housing Job_Type / PARAM = REF;
    MODEL Default_Risk(EVENT = '0') =
        Checking_Status Credit_History Duration_Months
        Savings_Amount Age_Years Credit_Amount
        Employment_Duration Installment_Rate
        Personal_Status_Sex Other_Debtors
        Residence_Duration Property
        Other_Installment_Plans Housing
        Number_Credits Job_Type
        Liable_People Telephone Foreign_Worker
        / SELECTION = STEPWISE
        SLENTY = 0.05
        SLSTAY = 0.05;
    SCORE DATA = work.test_data OUT = work.scored_test;
RUN;

```

/* 9. Running the Model on Test Data */

```

PROC LOGISTIC DATA = work.scored_test PLOT(ONLY) = ROC;
    MODEL Default_Risk(EVENT='0') = P_0;
    TITLE "Final Result";
RUN;

```

/*10. Generate the Classification Table to check Sensitivity */

```

PROC LOGISTIC DATA = work.scored_test;
    MODEL Default_Risk(EVENT='0') = P_0
        / CTABLE PPROB= 0.25;
    TITLE "Test 1: Sensitivity Analysis";
RUN;

```